

ODE

$$y' + py = Q$$

$$a_2 y'' + a_1 y' + a_0 y = f(x)$$

$$f(x) = 0$$

$$f(x) = ce^{\beta x} \Rightarrow y_p(x)$$

$$\begin{cases} c \cos \beta x, c \sin \beta x \\ p_n(x) \\ e^{\beta x} p_n(x) \end{cases} \quad a_2 y_p''(x) + a_1 y_p'(x) + a_0 y_p(x) = f(x)$$

- ① Laplace Transform ✓
- ② Introduce PDE ✓
- ③ chapter 13, Separation of variables
Review of Fourier Series ✓
- ④ chapter 12, linear PDE

Laplace Transform

$$f(t) \quad -\infty < t < \infty,$$

$$F(s) = \int_0^{\infty} f(t) e^{-st} dt \quad \checkmark$$

$p \rightarrow s$
complex
number

$$f(t) \xleftrightarrow{\quad} F(s)$$

↑
time

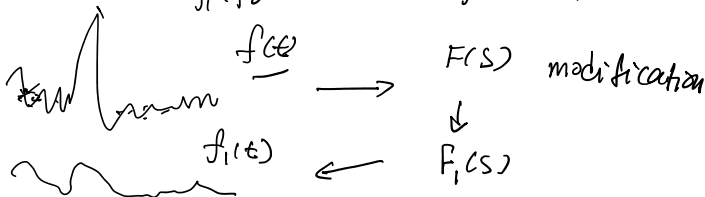
↑
frequency

$$s = \sigma + j\omega$$

$$F(s) = \int_0^{\infty} f(t) \underbrace{e^{-\sigma t}}_{\sigma=0} e^{-j\omega t} dt$$

$$F(\omega) = \int_{t_1+t_2}^{\infty} f(t) e^{-j\omega t} dt \quad \text{Fourier Transform}$$

$$f(t) = 0, t < 0$$



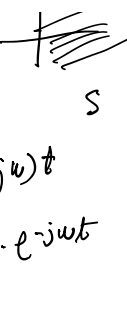
Ex $f(t) = 1$

$$F(s) = \int_0^{\infty} f(t) e^{-st} dt$$

$$= \int_0^{\infty} 1 \cdot e^{-st} dt$$

$$\underline{\text{Re}\{s\} > 0}$$

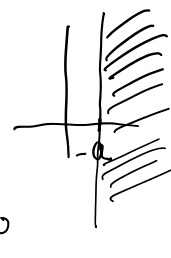
$$\begin{aligned}
 &= \int_0^{\infty} 1 \cdot e^{-st} dt \\
 &= -\frac{1}{s} e^{-st} \Big|_0^{\infty} \\
 &= -\frac{1}{s} [0 - 1] \\
 &= \frac{1}{s}
 \end{aligned}$$

$\text{Re}\{s\} > 0$ 

$$1 \Rightarrow \frac{1}{s}$$

$$\Sigma_{x^2} e^{-at} u(t)$$

$$\begin{aligned}
 F(s) &= \int_0^{\infty} e^{-at} \cdot e^{-st} dt \\
 &= -\frac{1}{(s+a)} e^{-(s+a)t} \Big|_0^{\infty} \\
 &= \frac{1}{s+a} \quad \text{Re}\{s+a\} > 0
 \end{aligned}$$



$$e^{-at} \Leftrightarrow \frac{1}{s+a}$$

$$\underline{e^{3t} + e^{-2t}} \Leftrightarrow \underline{\frac{1}{s-2} + \frac{1}{s+3}} = \frac{2s-1}{s^2-s-6}$$

P469

Property of Laplace Transform

$$f(t) \Leftrightarrow F(s) = \int_0^{\infty} f(t) e^{-st} dt$$

$$\begin{aligned}
 f'(t) \Leftrightarrow G(s) &= \int_0^{\infty} f'(t) e^{-st} dt \\
 &= \int_0^{\infty} e^{-st} df(t) \\
 &= \left[e^{-st} f(t) \right]_0^{\infty} - \int_0^{\infty} f(t) (-s) e^{-st} dt \\
 &= -s \int_0^{\infty} f(t) e^{-st} dt - f(0)
 \end{aligned}$$

$$f(t) \Leftrightarrow F(s) \quad \text{Differential}$$

$$f'(t) \Leftrightarrow s F(s) - f(0) \quad \text{property}$$

$$y(t) \Leftrightarrow Y(s)$$

$$g(t) = \underline{y'(t) \Leftrightarrow s Y(s) - y(0)} = G(s)$$

$$y''(t) = g'(t)$$

$$g'(t) \Leftrightarrow s G(s) - g(0)$$

$$[f'(t) u(t)] \Leftrightarrow s F(s) - f(0)$$

$$g'(t) \Rightarrow s \cdot G(s) - g(0) \\ = s[sY(s) - y(0)] - y'(0)$$

$$y''(t) \Rightarrow \underline{s^2 Y(s) - s y(0) - y'(0)} \quad \times s$$

$$y'''(t) \Rightarrow \underbrace{s^3 Y(s) - s^2 y(0) - s y'(0) - y''(0)}_{\text{initial conditions}}$$

Ex $y'' + 4y' + 4y = t^2 e^{-2t}, \quad y(0) = 0, \quad y'(0) = 0$

$$\begin{array}{c} -2, -2 \\ c_1 e^{-2t} + c_2 t e^{-2t} + (c_2 t^2 + a_2 t + a_3) e^{-2t} \end{array}$$

$$\begin{aligned} s^2 Y(s) + 4s Y(s) + 4Y(s) &= \frac{2}{(s+2)^3} && \text{transform} \\ (s^2 + 4s + 4) Y(s) &= \frac{2}{(s+2)^3} && \text{approach} \end{aligned}$$

$$Y(s) = \frac{2}{(s+2)^3 (s+2)^2}$$

$$y(t) = \frac{t^4 e^{-2t}}{12}$$

$$\underline{y''(t) + 2y'(t) + 3y(t) = \sin st}$$

$$\downarrow \\ \underline{[s^2 + 2s + 3] Y(s) = \boxed{}}$$

Ex 3 $y'' + 4y' + 13y = 20e^{-t}, \quad y(0) = 1, \quad y'(0) = 3$

$$\begin{aligned} &\downarrow \\ &+ 13Y(s) \\ &+ 4[sY(s) - 1] \\ &+ s^2 Y(s) - s - 3 = 20 \cdot \frac{1}{s+1} \end{aligned}$$

$$Y(s)[s^2 + 4s + 13] - s - 7 = \frac{20}{s+1}$$

$$\begin{aligned} Y(s)[s^2 + 4s + 13] &= \frac{20}{s+1} + s + 7 \\ &= \frac{s^2 + 8s + 27}{s+1} \end{aligned}$$

$$Y(s) = \frac{s^2 + 8s + 27}{(s+1)(s^2 + 4s + 13)} \quad \frac{k_3}{(s+\lambda_1)^2}$$

$$= \frac{k_1}{s+1} + \frac{k_2}{s+\lambda_1} + \frac{k_3}{s+\lambda_2} \quad \begin{array}{l} \nearrow \\ \text{if repeated} \end{array}$$

$$e^{-at} \Rightarrow \frac{1}{s+a}$$

$$e^{-at} \Rightarrow \frac{1}{s+a}$$

$$y(t) = k_1 e^{-t} + k_2 e^{-\lambda_1 t} + k_3 e^{-\lambda_2 t}$$

Fourier Series Harmonic

$$\cos \omega_0 t \quad \omega_0$$

$$\sin t \quad 2\pi$$


$T_0 = 2\pi$ fundamental period
周期用 2π

$$\sin 2t \quad T_0 = \pi$$

$$\sin \pi t \quad T_0 = 2$$

$$\sin \omega_0 t$$

$$\omega_0 T_0 = 2\pi$$

$$T_0 = \frac{2\pi}{\omega_0}$$

$$\sin \omega_0 t \quad \omega_0 \text{ radian frequency}$$

$$\omega_0 = 2\pi / T_0 \quad f_0 = 1/T_0 \text{ frequency in Hz}$$

$$\omega_0 = 2\pi f_0$$

$$\text{wifi: } 2.4 \text{ GHz}$$

$$f_0 =$$

$$\omega_0 = 2\pi / T_0$$

$$\sin \omega_0 t, \cos \omega_0 t$$

$$T_0 = 2\pi / \omega_0$$

$$\left\{ \begin{array}{lll} \sin \omega_0 t & \sin 2\omega_0 t & \sin 3\omega_0 t \dots \\ \cos \omega_0 t & \cos 2\omega_0 t & \cos 3\omega_0 t \dots \end{array} \right\}$$

Set of functions $\Rightarrow$ Basis functions orthogonal $\mathcal{L}_2$: $\langle f, g \rangle = \int_a^b f \cdot g \, dt$

$$f(t) = \sum_{n=0}^{\infty} C_n \phi_n(t)$$

$$= C_0 + \sum_{n=1}^{\infty} a_n \cos n\omega_0 t + \sum_{n=1}^{\infty} b_n \sin n\omega_0 t$$

$$\sin \frac{n\pi}{l} x$$

$$\omega_0 = n\pi / l = \frac{2\pi}{T_0}$$

$$T_0 = 2\pi / n\pi = 2l/n$$

$$\int_0^{T_0} \phi_n(t) \phi_m(t) \, dt = 0 \quad n \neq m$$



product
inner product

$$\int_0^{T_0} \phi_n(t) \phi_m(t) \, dt = 0 \quad n \neq m$$

$$\int_0^{T_0} \cos n \omega_0 t \cos m \omega_0 t dt = 0 \quad n \neq m$$

$$= T_0/2 \quad n = m$$

$$\int_0^{T_0} \sin n \omega_0 t \sin m \omega_0 t dt = 0 \quad n \neq m$$

$$= T_0/2 \quad n = m$$

$$\int_0^{T_0} \sin n \omega_0 t \cos m \omega_0 t dt = 0$$

$$\int_0^{T_0} \cos n \omega_0 t f(t) dt = \int_0^{T_0} \left[C_0 + \sum_{n=1}^{\infty} \left(\frac{a_n}{2} \cos n \omega_0 t + \frac{b_n}{2} \sin n \omega_0 t \right) \right] f(t) dt$$

$$\int_0^{T_0} \cos 2 \omega_0 t f(t) dt = a_2 \cdot T_0/2$$

$$a_2 = \frac{2}{T_0} \int_0^{T_0} f(t) \cos 2 \omega_0 t dt$$

$$a_n = \frac{2}{T_0} \int_0^{T_0} f(t) \cos n \omega_0 t dt$$

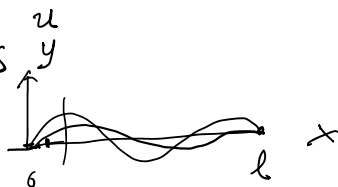
$$b_n = \frac{2}{T_0} \int_0^{T_0} f(t) \sin n \omega_0 t dt$$

$$C_0 = \frac{1}{T_0} \int_0^{T_0} f(t) dt$$

$$f(t) = C_0 + \sum_{n=1}^{\infty} \left(\frac{a_n}{2} \cos n \omega_0 t + \frac{b_n}{2} \sin n \omega_0 t \right)$$

Fourier Series

Separation of variables



$u(x,t)$

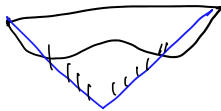
wave equation

$$u_{tt} - a^2 u_{xx} = 0 \quad 0 \leq x < l \quad t > 0$$

$$u|_{x=0} = 0 \quad u|_{x=l} = 0 \quad \text{edge condition}$$

$$\begin{cases} u|_{x=0} = 0 & u|_{x=l} = 0 & \text{edge condition} \\ u|_{t=0} = \phi(x) & & \text{initial condition} \\ u_t = \psi(x) & & \end{cases}$$

两端固定自由振动



Step 1 $u(x, t) = X(x) T(t)$

$$T'(t) X(x) - a^2 X''(x) T(t) = 0$$

$$\frac{1}{a^2} \frac{T'(t)}{T(t)} = \frac{X''(x)}{X(x)} = -\lambda$$

$$\begin{cases} X''(x) + \lambda X(x) = 0 & \textcircled{A} \checkmark \\ T''(t) + \lambda a^2 T(t) = 0 & \textcircled{B} \end{cases}$$

$$X(x) T(t) \Big|_{x=0} = X(0) T(t) = 0 \Rightarrow X(0) = 0$$

$$X(x) T(t) \Big|_{x=l} = X(l) T(t) = 0 \Rightarrow X(l) = 0 \quad \checkmark$$

Step 2

$$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(l) = 0 \end{cases}$$

① $\lambda = 0$ $X''(x) = 0$ $X(x) = Ax + B$
 $A = 0$ $B = 0$

$$X(x) \equiv 0$$

$$u(x, t) = X(x) T(t) = 0$$

Not True

② $\lambda < 0$

$$X(x) = A e^{\sqrt{\lambda} x} + B e^{-\sqrt{\lambda} x}$$

$$X(x) \equiv 0$$

$$X''(x) - 4X(x) = 0$$

$$e^{2x} \quad e^{-2x}$$

$$X = A e^{2x} + B e^{-2x}$$

$$X(0) = A + B = 0$$

$$X(l) = A e^{2l} + B e^{-2l} = 0$$

$$\left. \begin{matrix} A = 0 \\ B = 0 \end{matrix} \right\}$$

Not True!

③ $\lambda > 0$ has to be positive!!

$$\begin{vmatrix} 1 & 1 \\ e^{2l} & e^{-2l} \end{vmatrix} \neq 0$$

determinant 行列式

eigen function $\left\{ \begin{array}{l} X''(x) + \lambda X(x) = 0 \\ X(0) = X(l) = 0 \end{array} \right.$

$$\frac{e^{i\sqrt{\lambda}x}}{e^{-i\sqrt{\lambda}x}}$$

$X(x) = A \cos \sqrt{\lambda} x + B \sin \sqrt{\lambda} x$

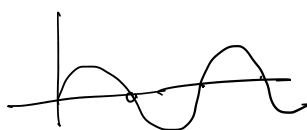
$$X(0) = A \cdot 1 + B \cdot 0 = 0 \Rightarrow A = 0$$

$$X(l) = B \sin \sqrt{\lambda} l = 0 \quad \underline{B \neq 0}$$

$$\sin \sqrt{\lambda} l = 0 \quad \text{Must!}$$

$$\sqrt{\lambda} l = n\pi \quad n=1, 2, 3, 4$$

$$\sqrt{\lambda} = \frac{n\pi}{l}$$



eigen function $X_n(x) = \sin \frac{n\pi}{l} x$

$n=1, 2, 3, 4 \dots$
非负整数

λ : eigen value 特征值.

$\lambda = \frac{n\pi^2}{l^2}$ $n=1, 2, 3, \dots$

$$\sqrt{\lambda} a$$

step 3

$$T''(t) + \lambda a^2 T(t) = 0$$

$$T_n(t) = C_n \cos \sqrt{\lambda} a t + D_n \sin \sqrt{\lambda} a t$$

$$= C_n \cos \frac{n\pi}{l} a t + D_n \sin \frac{n\pi}{l} a t$$

$n=1, 2, 3, \dots$

step 4

$$u_n(x, t) = \sin \frac{n\pi}{l} x \left(C_n \cos \frac{n\pi}{l} a t + D_n \sin \frac{n\pi}{l} a t \right) \star$$

7) xibai tui

$$(u_n)_{tt} = a^2 (u_n)_{xx}$$

$$T_n''(t) X_n(x)$$

//

$$\begin{aligned}
 (u_n)_{tt} &= a^2 (u_n)_{xx} \quad (u_n(t, x)) \\
 &= T_n''(t) X_n(x) - a^2 T_n(t) X_n''(x) \\
 &= -\lambda a^2 X_n(x) - a^2 T_n(t) [-\lambda X_n(x)] \\
 &= 0
 \end{aligned}$$

$$\sum_{n=1}^{\infty} u_n(x, t) = u(x, t) \quad u_{tt} - a^2 u_{xx} = 0 \quad \checkmark$$

$$u|_{x=0} = u|_{x=l} = 0 \quad \checkmark$$

$$u(x, t) = \sum_{n=1}^{\infty} \sin \frac{n\pi}{l} x \left(C_n \sin \frac{n\pi}{l} at + D_n \cos \frac{n\pi}{l} at \right)$$

$$\sin n\omega_0 t$$

$$u|_{t=0} = \varphi(x)$$

$$\sum_{n=1}^{\infty} D_n \sin \frac{n\pi}{l} x = \varphi(x) \quad \omega_0 = \frac{\pi}{l} = \frac{2\pi}{T_0}$$

$$D_n = \frac{2}{T_0} \int_0^{T_0} \varphi(x) \sin \frac{n\pi}{l} x \quad (T_0 = \frac{2\pi}{\omega_0} = \frac{2\pi l}{\pi})$$

$$\boxed{D_n = \frac{2}{2l} \int_0^{2l} \varphi(x) \sin \frac{n\pi}{l} x dx} = 2l$$

$$= \frac{1}{l} \int_0^{2l} \varphi(x) \sin \frac{n\pi}{l} x dx$$

$$u_t|_{t=0} = \psi(x)$$

$$u(x, t) = \sum_{n=1}^{\infty} \sin \frac{n\pi}{l} x \left(C_n \frac{n\pi a}{l} \cos \frac{n\pi}{l} t + \underbrace{D_n \left(-\frac{n\pi}{l} \right) \sin \frac{n\pi}{l} t}_{t=0} \right)$$

$$= \psi(x)$$

$$\psi(x) = \sum_{n=1}^{\infty} \sin \frac{n\pi}{l} x \cdot C_n \frac{n\pi a}{l}$$

$$C_n \frac{n\pi a}{l} = \frac{2}{T_0} \int_0^{T_0} \psi(x) \sin \frac{n\pi}{l} x \, dx$$

$$\underline{C_n} = \frac{2}{2n\pi a} \int_0^{2l} \psi(x) \sin \frac{n\pi}{l} x \, dx$$

$$u(x, t) = \sum_{n=1}^{\infty} \sin \frac{n\pi}{l} x \left[C_n \sin \frac{n\pi}{l} t + D_n \cos \frac{n\pi}{l} t \right]$$

$\phi(x)$

$\psi(x)$

$$\int_0^l X_n(x) X_m(x) dx = 0$$

$$\left\{ \begin{array}{l} \int_0^l \sin \frac{n\pi}{l} x \sin \frac{m\pi}{l} x \, dx = 0 \\ \int_0^l \left[\sin \frac{n\pi}{l} x \right]^2 dx = \frac{l}{2} \end{array} \right.$$